4. Show that $k(A,B) = |A \cap B|\frac{1}{|A \cup B|}$ is p.d.

Using the shorthand $\sum \equiv \sum\limits_{k}^{n}$ we procede as follows:
For the case that $k(A,B) = |A \cap B|\frac{1}{|A \cup B|}$ we assume that $A \cup B \neq \emptyset$

To show that K is p.d it is suffient to show that $\frac{1}{|A \cup B|}$ is p.d. since have shown that
$|A \cap B|$ is p.d. earlier. Then we can write

$$k(A,B) = \frac{\sum \mathbb{1}_A \cdot \mathbb{1}_B}{\sum \left(\mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_A \cdot \mathbb{1}_B\right)} = D\left(\frac{1}{1-D}\right)$$

with $D \equiv \dfrac{\sum \mathbb{1}_A \cdot \mathbb{1}_B}{\sum (\mathbb{1}_A + \mathbb{1}_B)}$

$A \cup B \neq \emptyset$ implies that at least one $x_i \in A$ or $x_i \in B$.

case 1: All the points are in either A and not B or visa vera. Then
$D = 0 \implies k(A,B) = 0$

case 2: All the points are in $A \cap B \implies D = 2\dfrac{n}{(n+n)} = 1 \implies k(A,b) = 1$

case 3: D is such that $0 < D < 1$.

In this case we can make use of the geometric series which allows D to be expressed as

$$k = D\left(\frac{1}{1-D}\right) = D\left(1 + D + D^2 + \cdots\right).$$

Since the product of p.d. kernels is also p.d, it is sufficient to show that D is p.d. In fact, since $D = D_1 D_2$ with $D_1 = \sum \mathbb{1}_A \cdot \mathbb{1}_B$. We need only to show that $D_2 \equiv \dfrac{1}{\sum (\mathbb{1}_A + \mathbb{1}_B)}$

To this end we will try to express D_2 as an function of the intersection between A and B and see if it represents an inner product between A and B.